# Placard main board

Hierarchy: each box is one sheet; the sheets connect through global labels

### Power and clock

J13 USB-C, F1, U40 3.3 V

X1 25.175 MHz clock

STALL / RUN jumpers

File: power.kicad_sch

### Control

U1 State, U2 Branch

U17 instruction register PLD

File: control.kicad_sch

### Program counter and address mux

U3, U4 program counter

U7–U10 address mux

File: pc_mux.kicad_sch

### Memory and I/O decode

U31, U32 RAM, J8 cartridge

U11 memory control

U12 I/O decode, U25 map

File: memory.kicad_sch

### Video timing

U13 H, U14 V counters

U15 Sync, U35 audio slots

U29 line read

File: video_timing.kicad_sch

### ALU and registers

U5, U6 ALU

U19 AC, U20 X, U21 Y

File: alu_regs.kicad_sch

### Data bus

U18 D register

U26, U27 source drivers

R5–R12, RN1

File: data_bus.kicad_sch

### Expansion slots

J9–J12 card-edge slots

File: expansion.kicad_sch

### Video and audio out

U22 pixel buffer, U23 line code

U16 color decoder, VGA

U36 audio DAC, J6

File: video_out.kicad_sch

### Shared port

U38 port PLD, U42 PS/2 receiver

J2–J5

File: shared_port.kicad_sch

34 chips + U40 regulator (docs/CHIPS.md numbers; U24, U28, U30, U33, U34 are not used).
Decoupling: every chip's 100 nF Cn sits beside Un on its own sheet.
Power: +5V, GND, +3V3 (SD card), GNDA (audio, joined to GND on sheet 2).
The ROM cartridges are their own projects: kicad/cartridge_64k/, kicad/cartridge_32k/.

Chip numbers U1–U42 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md

**Placard computer**

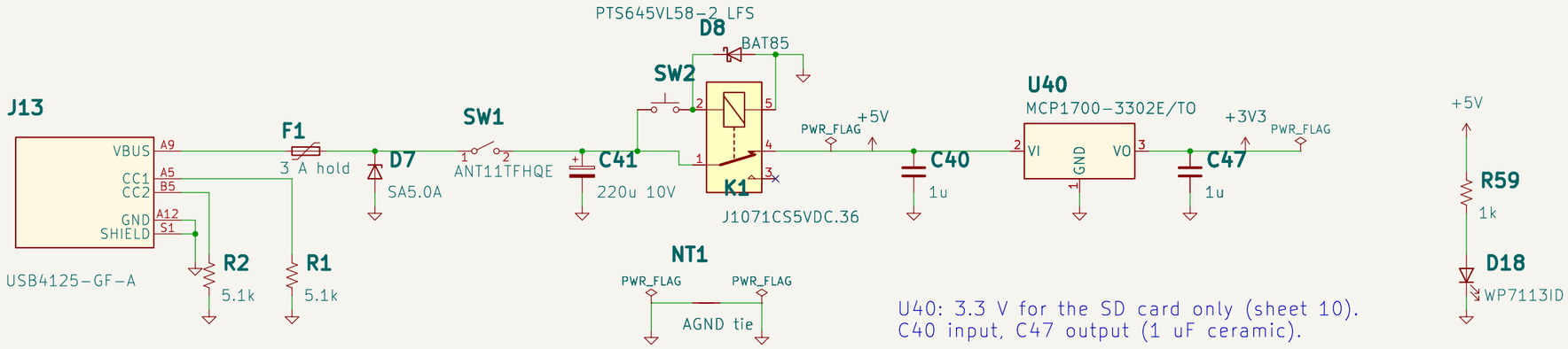
Sheet: /
File: placard.kicad_sch

***Title: Placard: Main board***

| Size: A3 | Date: 2026-10-26 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. | kicad-cli 7.0.11+dfsg-1build4 | Id: 1/11 |

# Power and clock

USB-C 5 V in, 3.3 V for the SD card, 25.175 MHz oscillator (CLK), cartridge STALL and RUN



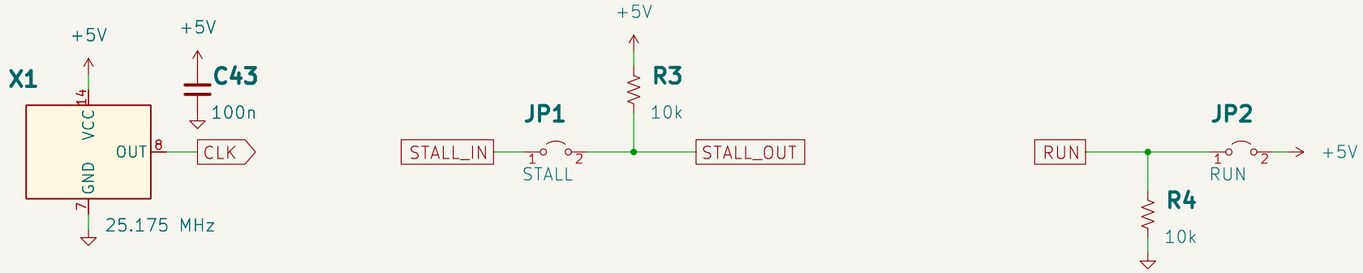

5.1k on CC1 and CC2: a USB-C source gives 5 V. 19 PLDs draw up to about 2.5 A (130 mA each at −7, worst case): use a 3 A supply; the polyfuse trips on a short. SW1 power: toggle, common + one throw. SW2 reset: holding it pulls in K1, whose NC contact opens the 5 V line.

U40: 3.3 V for the SD card only (sheet 10). C40 input, C47 output (1 uF ceramic).

AGND (audio) meets GND here, at the power input




D18 power LED (red, 5 mm) on +5V, after K1: dark while reset is held. R59 1k: about 3 mA.




STALL_OUT (10k up) and RUN (10k down) come from the cartridge: STALL_IN (the Branch PLD's STALL) goes out to it. Without one, fit JP1 and JP2 for bring-up (BRINGUP.md): take them out with a cartridge.

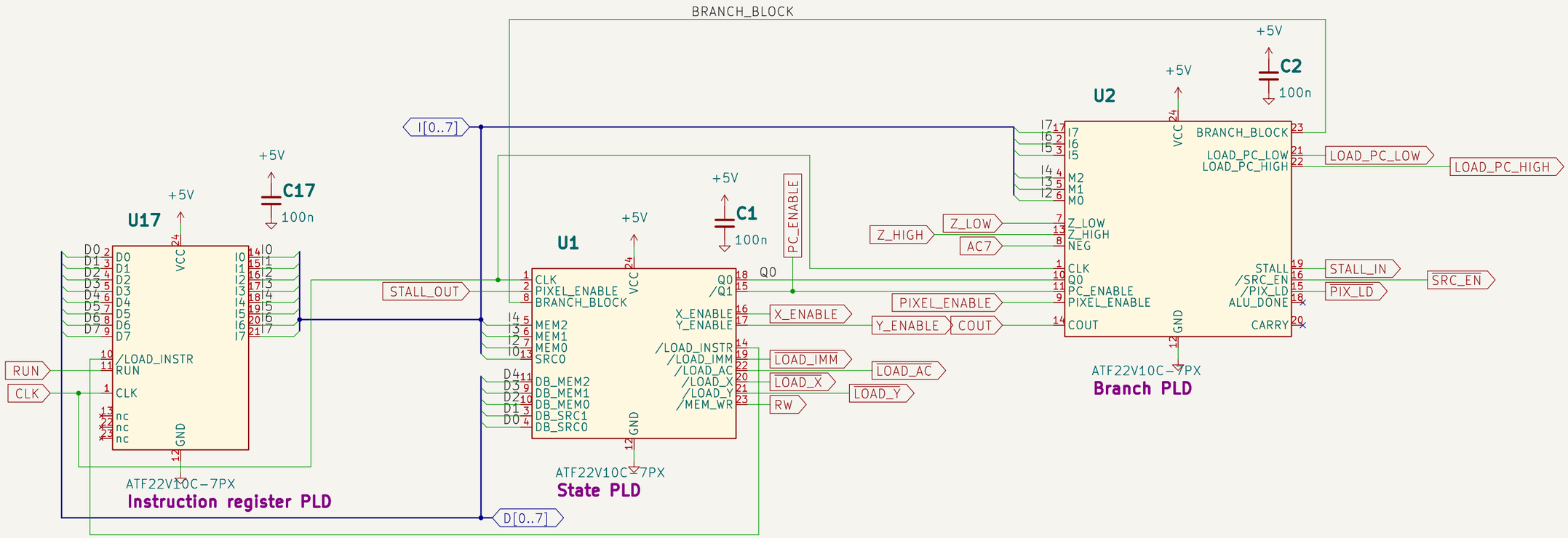
Chip numbers U1–U42 = docs/CHIPS.md; decoupling Cn goes with Un
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**Placard computer**

| | | |
|---|---|---|
| Sheet: /Power and clock/<br>File: power.kicad_sch | | |
| ***Title: Placard: Power and clock*** | | |
| Size: A3 | Date: 2026-10-26 | **Rev: 1** |
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# Control

State and Branch PLDs, instruction register



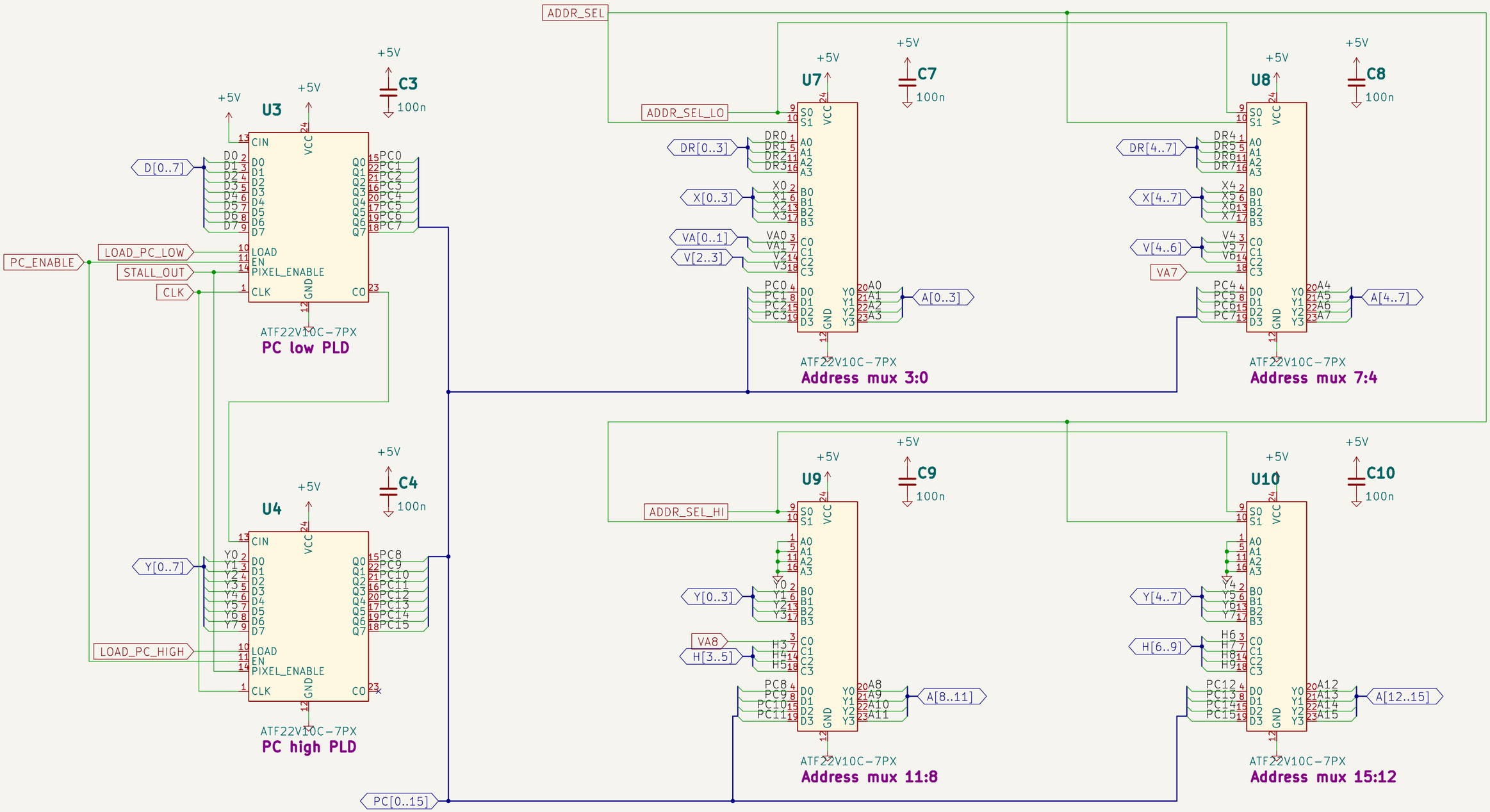

U17 loads the opcode in FETCH (/LOAD_INSTR) and holds 02h while RUN is low.
U2's STALL (STALL_IN) goes to the cartridge; STALL_OUT comes back to U1 and the PC.
U1 State: FETCH / LOAD / EXEC, register loads (active low, F377 /E), RW (1 = read).
U2 Branch: NEG is AC7.  ALU_DONE (18) and CARRY (20) are internal registers: no connection.
Bring-up straps S1-S3 go in U1's socket (BRINGUP.md), not on the board.

Chip numbers U1-U42 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

| | | |
|---|---|---|
| Sheet: /Control/ | | |
| File: control.kicad_sch | | |
| ***Title: Placard: Control*** | | |
| Size: A3 | Date: 2026-10-26 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 3/11 |

# Program counter and address mux

Mux inputs: A = D / 0, B = X / Y, C = video address, D = PC




Video address: VA0, VA1, VA7, VA8 from the audio slot PLD (line repeat), A6:A2 = V6:V2, A15:A9 = H9:H3.
U3 CIN tied high (always counts); U3 CO carries into U4. STALL_OUT holds the PC (video reads, ALU wait, cartridge load).

Chip numbers U1–U42 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md

**Placard computer**

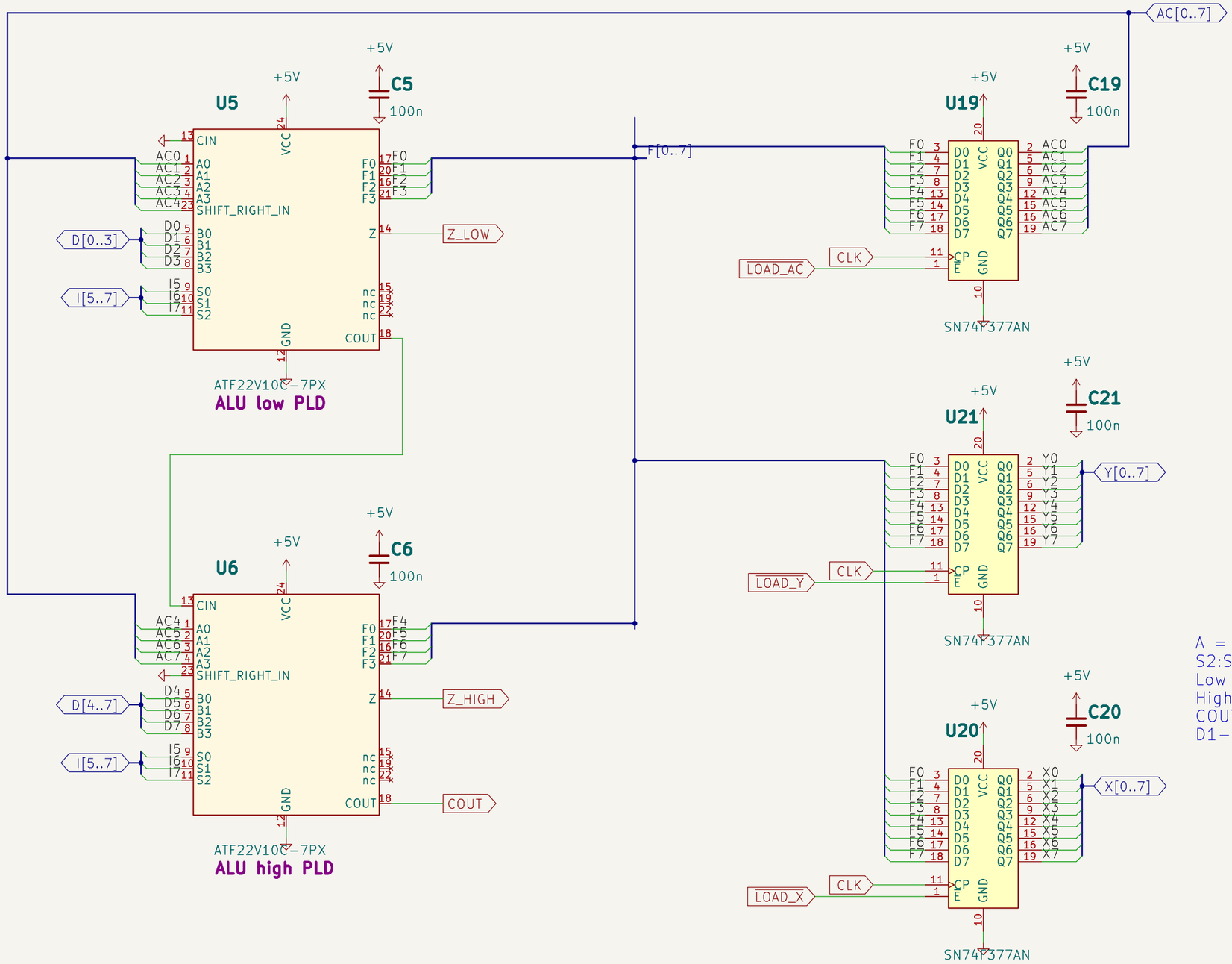
Sheet: /Program counter and address mux/
File: pc_mux.kicad_sch

***Title: Placard: Program counter and address mux***

| Size: A3 | Date: 2026-10-26 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 4/11 |

# ALU and registers

Two 4-bit ALU PLDs; F7:F0 loads AC, X or Y



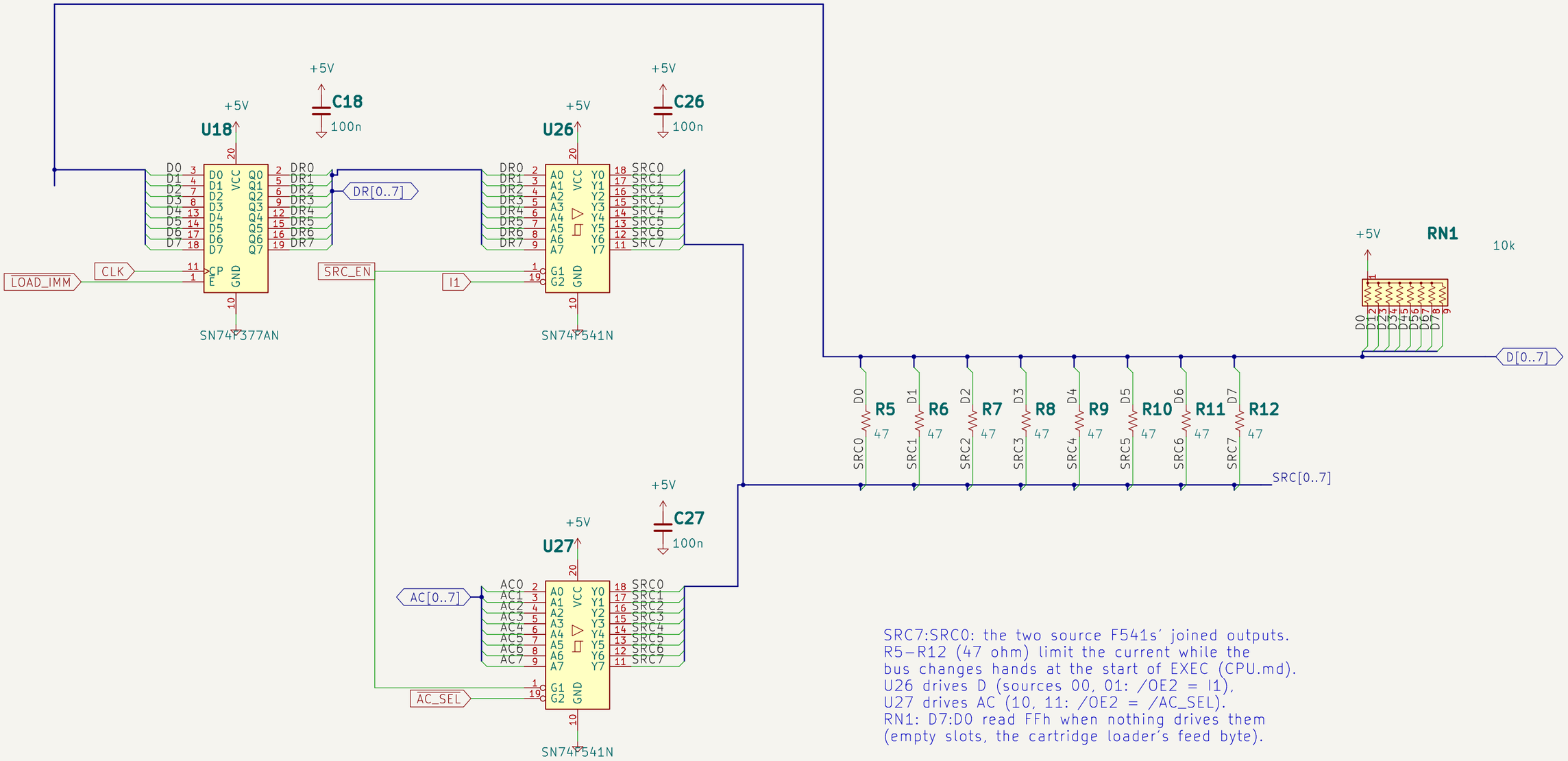

A = AC, B = the data bus (the source in EXEC).
S2:S0 = I7:I5.
Low nibble: CIN = 0, shift-right in = AC4.
High nibble: CIN = low COUT, shift-right in = 0.
COUT (high) is the carry for the Branch PLD.
D1–D3 (15, 19, 22) are internal carry nodes.

Chip numbers U1–U42 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

| | | |
|---|---|---|
| Sheet: /ALU and registers/ | | |
| File: alu_regs.kicad_sch | | |
| ***Title: Placard: ALU and registers*** | | |
| Size: A3 | Date: 2026-10-26 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 5/11 |

# Data bus

D register, source drivers (D or AC) onto the one data bus through series resistors, read pull-ups




SRC7:SRC0: the two source F541s' joined outputs.
R5–R12 (47 ohm) limit the current while the
bus changes hands at the start of EXEC (CPU.md).
U26 drives D (sources 00, 01: /OE2 = I1),
U27 drives AC (10, 11: /OE2 = /AC_SEL).
RN1: D7:D0 read FFh when nothing drives them
(empty slots, the cartridge loader's feed byte).

Chip numbers U1–U42 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md

**Placard computer**

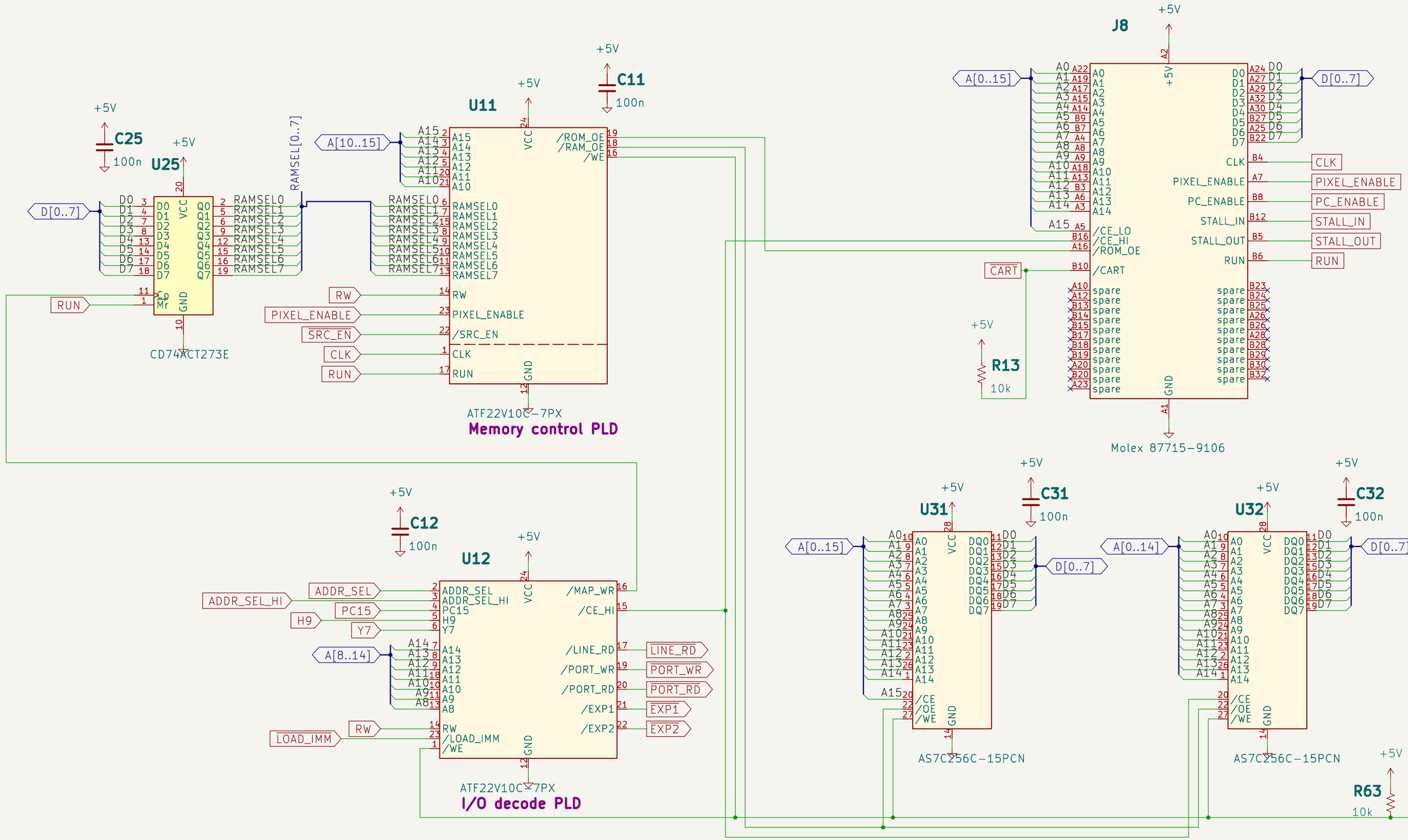
Sheet: /Data bus/
File: data_bus.kicad_sch

***Title: Placard: Data bus***

| Size: A3 | Date: 2026-10-26 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 6/11 |

# Memory and I/O decode

RAM, ROM cartridge slot, memory control, I/O decode, map register




Lower 32K: /CE = A15 (RAM U31, cartridge /CE_LO); upper: /CE_HI from U12.
RUN low (cartridge loading): /ROM_OE held off, map register clear (all ROM), /WE let go (R63).
/CART reads low with a cartridge in; nothing reads it yet.  21 spare contacts.

Chip numbers U1–U42 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md

**Placard computer**

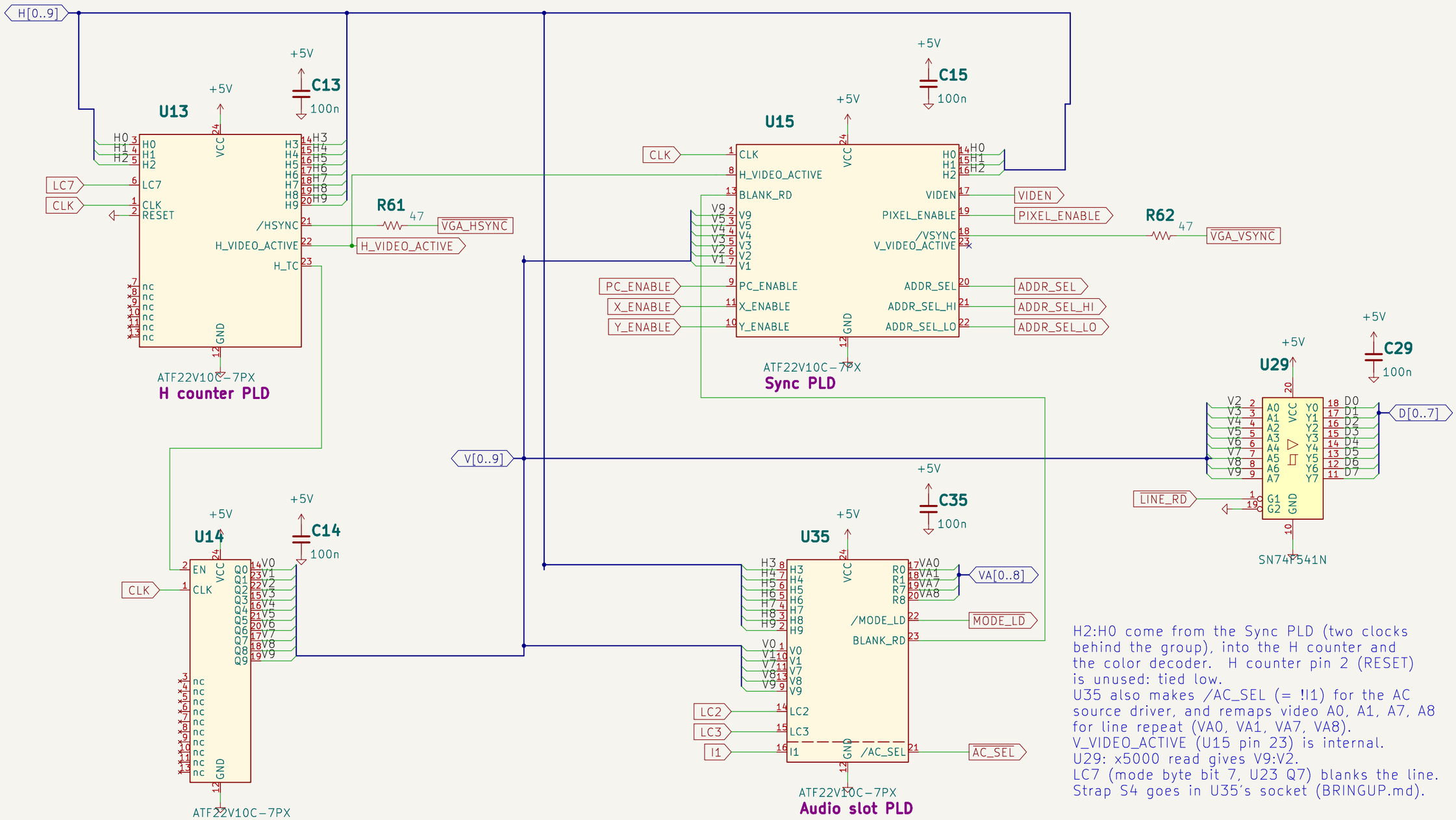
Sheet: /Memory and I/O decode/
File: memory.kicad_sch

***Title: Placard: Memory and I/O decode***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-26 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 7/11 |

# Video timing

640 x 480 at 60 Hz: H and V counters, Sync, mode and audio slots, line read



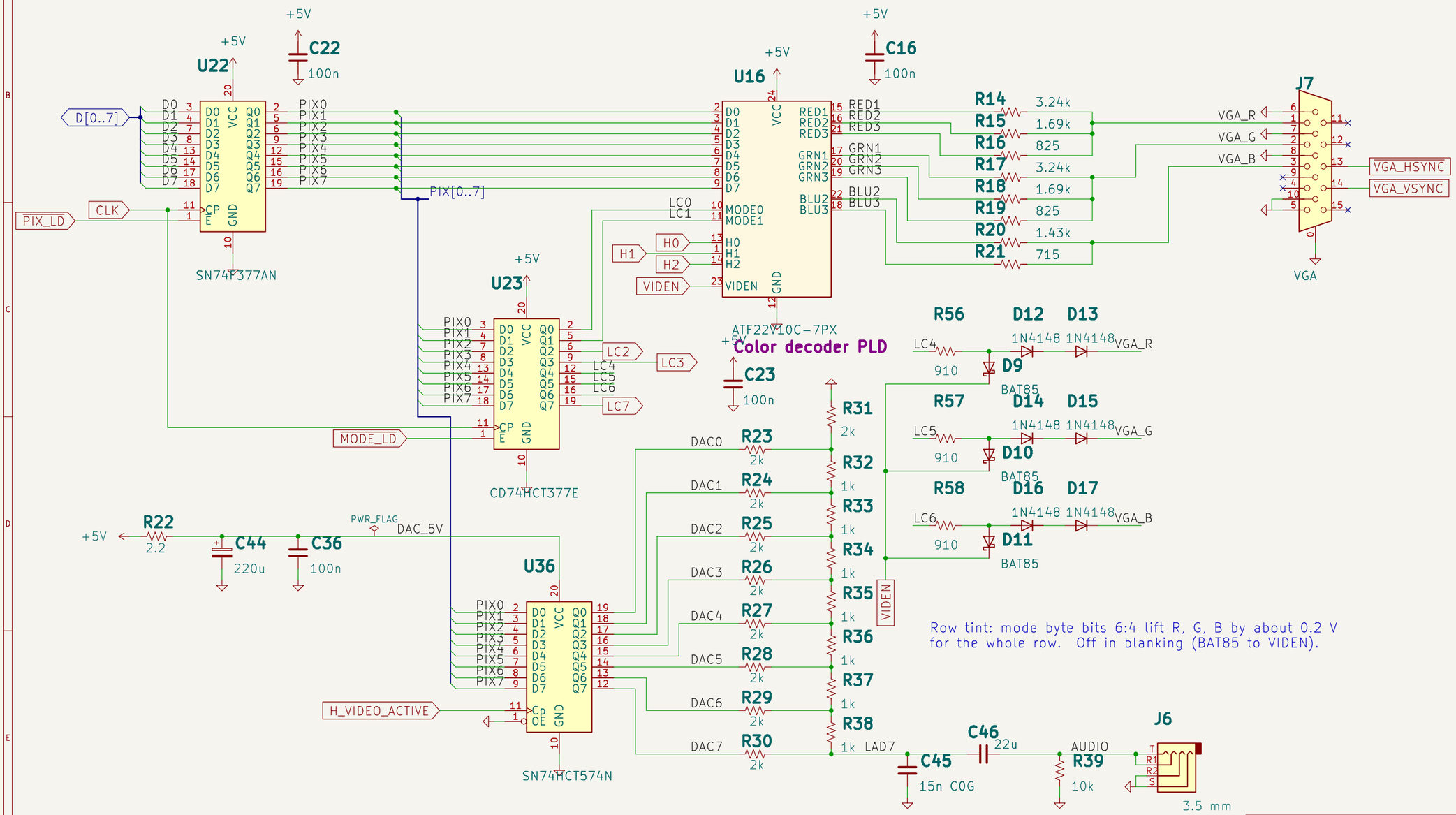

H2:H0 come from the Sync PLD (two clocks behind the group), into the H counter and the color decoder. H counter pin 2 (RESET) is unused: tied low.
U35 also makes /AC_SEL (= !I1) for the AC source driver, and remaps video A0, A1, A7, A8 for line repeat (VA0, VA1, VA7, VA8).
V_VIDEO_ACTIVE (U15 pin 23) is internal.
U29: x5000 read gives V9:V2.
LC7 (mode byte bit 7, U23 Q7) blanks the line.
Strap S4 goes in U35's socket (BRINGUP.md).

Chip numbers U1–U42 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

| | | |
|---|---|---|
| Sheet: /Video timing/ | | |
| File: video_timing.kicad_sch | | |
| ***Title: Placard: Video timing*** | | |
| Size: A3 | Date: 2026-10-26 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 8/11 |

# Video and audio out

Pixel buffer, line code, color decoder and VGA DAC; audio DAC register, R–2R ladder, audio line out




U23 Q1:Q0 = color mode; Q3:Q2 = line repeat to U35 and Q7 = blank line to U13 (sheet 8); Q6:Q4 = row tint.
DAC_5V: U36's own filtered rail (2.2 ohm, 220u, C36).
Ladder: 2k 1 % per bit + 2k termination, 1k between nodes (LAD7 = out). AGND (GNDA) joins GND at the power input.

Chip numbers U1–U42 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

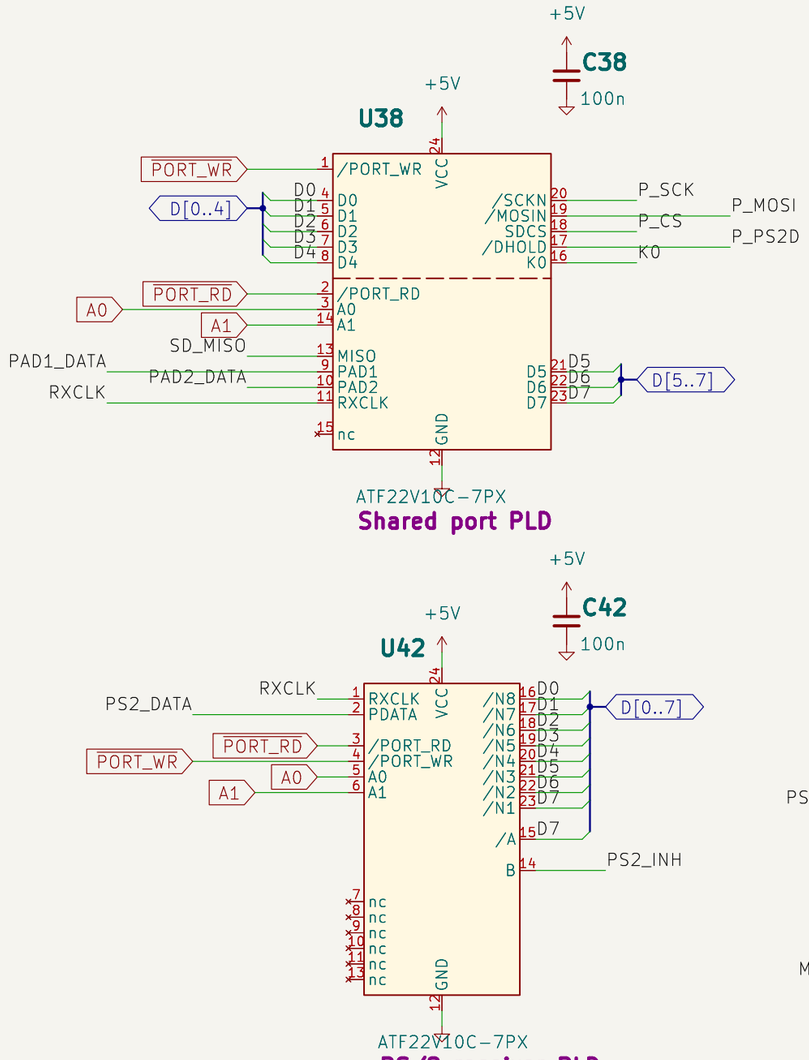
Sheet: /Video and audio out/
File: video_out.kicad_sch

***Title: Placard: Video and audio out***

| Size: A3 | Date: 2026-10-26 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 9/11 |

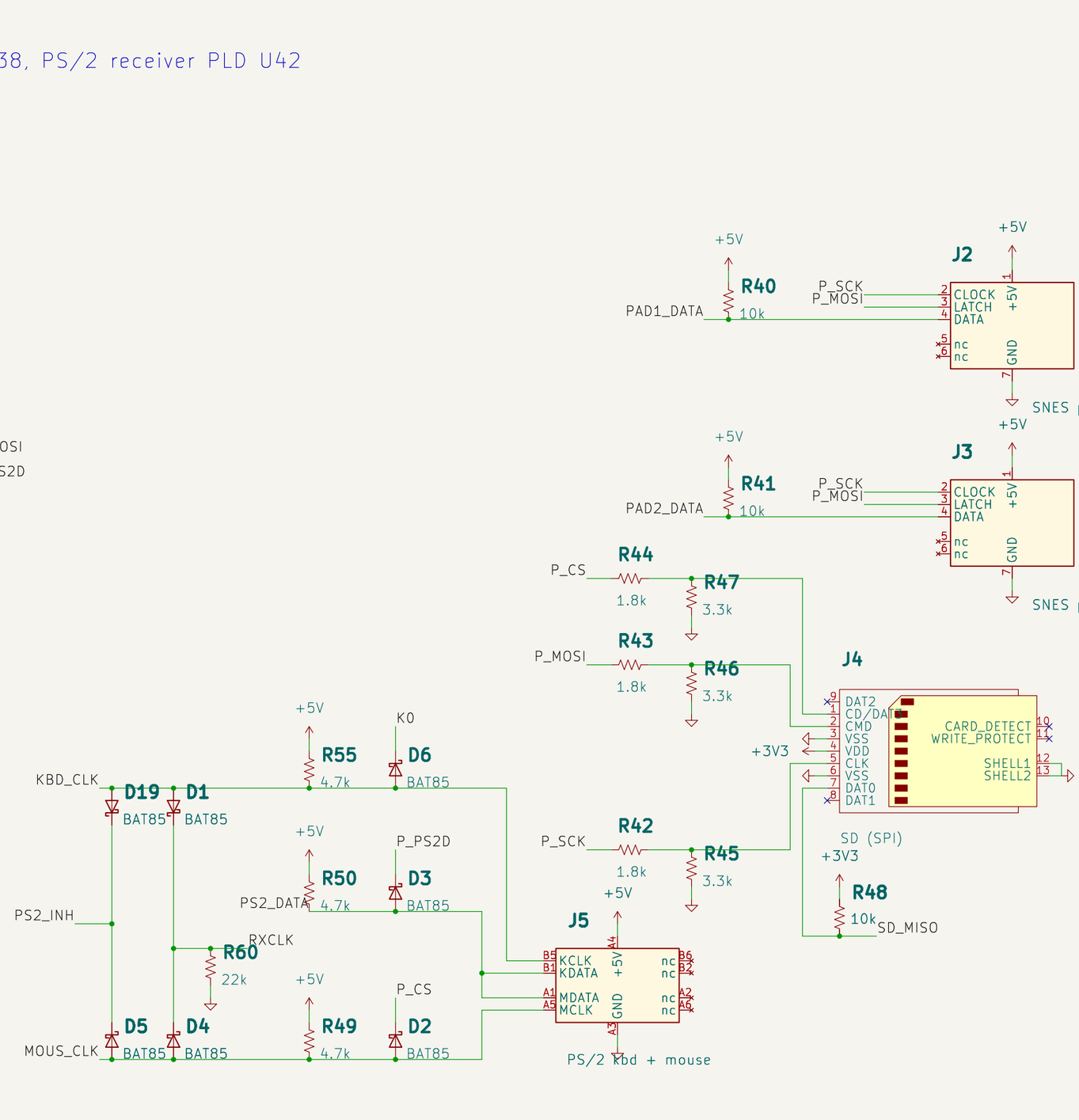
# Shared port x5200

Two SNES pads, SD card (SPI), PS/2 keyboard and mouse: port PLD U38, PS/2 receiver PLD U42









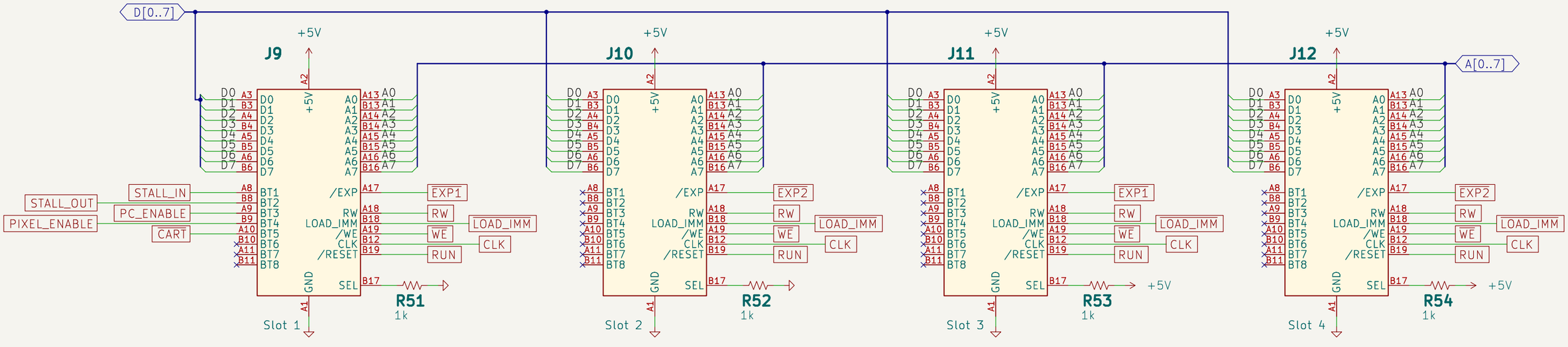

U38 register (rests at EB): bit 0 = SCK and pad CLOCK, bit 1 = MOSI and pad LATCH, bit 2 = SD /CS (low also holds the mouse CLK), bit 3 = PS/2 DATA hold, bit 4 = K0 (low holds the keyboard CLK). Reads x5200/x5201: D7 MISO or pad 1, D6 pad 2, D5 RXCLK. U42 clocks on RXCLK (the free device's clock through D1/D4, 22k down) and holds the byte: read at x5202, ready = x5203 D7 low; after the 11th clock it pulls both clocks low through D19/D5 until a write to x5202 re-arms it. PS/2 DATA is shared by both sockets. SD lines come down to 3.2 V through 1.8k / 3.3k; MISO (3.3 V) goes straight to the PLD's TTL input.

Chip numbers U1–U42 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

| | | |
|---|---|---|
| Sheet: /Shared port/ | | |
| File: shared_port.kicad_sch | | |
| ***Title: Placard: Shared port*** | | |
| Size: A3 | Date: 2026-10-26 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 10/11 |

# Expansion slots

Four 2 x 20 card-edge slots, x5100-x53FF, 128 bytes each (EXPANSION.md)




SEL: GND on slots 1 and 2, +5 V on slots 3 and 4 (through 1k): the A7 half the card answers.
/EXP1 goes to slots 1 and 3, /EXP2 to slots 2 and 4.  /RESET = RUN (low while the cartridge loads).
BT1-BT8 (boot contacts): slot 1 only, STALL_IN, STALL_OUT, PC_ENABLE, PIXEL_ENABLE, /CART; cards latch D7:D0.
LOAD_IMM carries the State PLD's /LOAD_IMM (low when the CPU latches the read bus into D).

Chip numbers U1-U42 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

Sheet: /Expansion slots/
File: expansion.kicad_sch

***Title: Placard: Expansion slots***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-26 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 11/11 |